

Electromagnetism

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1 Particle's charge

Every particle has an intrinsic property called charge. This is a numerical value Q which can be positive or negative. Some classic examples are electrons with negative charge and protons with positive charge, which is the reason why, chemistry-wise, we use e^- to refer to electrons and p^+ to refer to protons.

The charge has a meaningful behavior: opposite charges (positive and negative) attract each other and same charges (positive and positive, or negative and negative) repel each other.

This is key to understand everything that comes next as it is the fundamental idea of the whole topic.

These charges have the unit coulomb ($[Q] = C$).

1.1 Coulomb's law

When we take a look to when two particles are close to each other and they are either repelling or attracting each other, we can try to find out a formula that describe the force acting upon the particles. By doing some experimentation, we see that:

$$F \propto Q_1 \quad F \propto Q_2 \quad F \propto r^2$$

Where r is the distance between the two particles and F the magnitude of the vector force $F = \|\vec{F}\|$.

At this point, these proportions are pretty much the same as the gravitational force and, therefore, we will obtain a similar formula.

Mixing these pieces together, we have:

$$F = k \frac{Q_1 Q_2}{r^2}$$

And again, by doing some experiments, we can obtain the value of the constant k :

$$k = 8.99 \cdot 10^9 \frac{N \cdot m^2}{C^2} = \frac{1}{4\pi\epsilon_0} \frac{N \cdot m^2}{C^2}$$
$$\therefore F = \frac{Q_1 Q_2}{4\pi\epsilon_0 r^2} \implies \vec{F} = \frac{Q_1 Q_2}{4\pi\epsilon_0 r^2} \hat{r}$$

Where $\hat{r}$ is a unit vector pointing from one charge to the other.